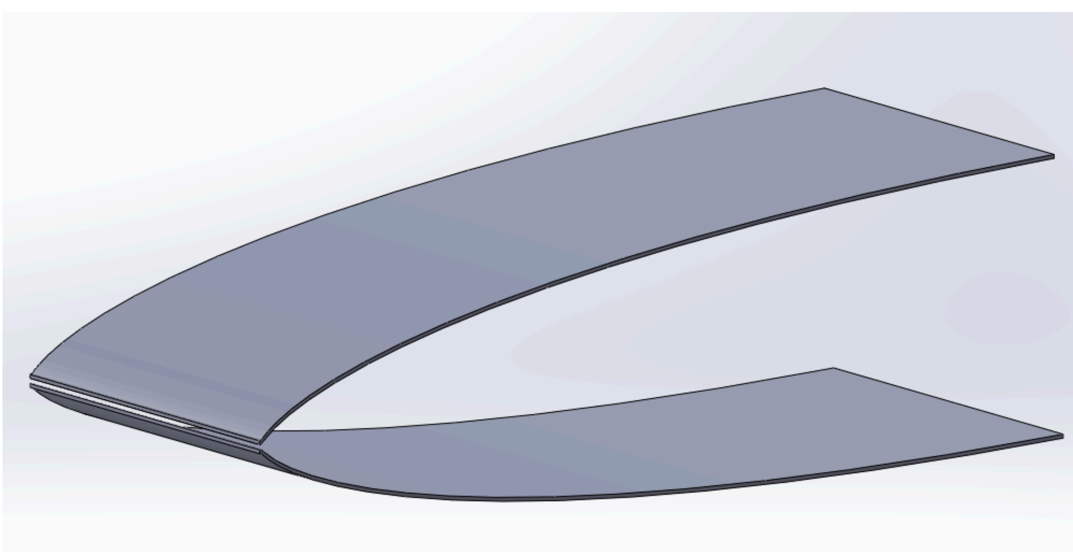
0.0 - Wind Tunnel Design

The end goal of this project was to design a wind tunnel with a test-section Mach number of 6. To accomplish this, the method of characteristics was employed, utilizing given design parameters in a MATLAB script to automate calculations. From the newly designed wind tunnel, the aerodynamic forces were found, and the tunnel run time was calculated.

The main constraints of the wind tunnel were as follows:

- The nozzle must be a planar 2-D nozzle with an exit Mach number of 6.
- The exit height is 350 mm and the nozzle width is 500 mm.
- The throat is not a sharp corner. The throat expansion region is a circular arc with radius of curvature equal to the throat half-height.
- The expansion through the throat is approximated with N equally spaced Prandtl-Meyer waves, and the code must be able to run for different values of N.
- The test-section back pressure is held at pb = 0.1 psia, and the pressure tank stores air at 2000 psia and 300 K.
- The free-stream static temperature must remain above 55 K so air does not liquefy.

Below is a SOLIDWORKS model given the nozzle contours, with an arbitrary thickness. Note that this model does not consider the required geometry upstream of the throat, but is only meant to serve as an illustration of the 2D planar nozzle geometry generated by the method-of-characteristics.



1.0 - How the Code Works

1.1 - About the method of characteristics

In supersonic flow, information travels along so-called characteristic lines at any given point. These lines are theoretical values with properties such as mach number and flow direction, and other state information. The flow properties actually change in a continuous field, but certain properties can be assumed along these so-called characteristic lines. This is used to discretize the flow region for computer calculations.

These characteristic lines have known relationships and properties. They reflect off walls, and can be reflected off other lines (in a symmetric nozzle, they reflect off the centerline where symmetric waves are assumed to meet), and they can intersect with other waves. The intersections and reflections of these characteristic lines dictate how the flow behaves.

The method of characteristics (abbreviated as "MOC") uses the reflection and intersection properties of the characteristic lines to calculate the local Prandtl-Meyer expansion angle ν, and flow angle at a given point or "node." The Prandtl-Meyer angle directly corresponds to a given mach number. The mach number can also be used to find other characteristics, such as the pressure or temperature, from other relations, namely, isentropic flow. Such is the fundamental working principle of the method of characteristics and is, at its core, what the code is doing.

1.2 - Initial Parameters

The code first calculates initial parameters based on the given flow and geometric parameters. The exit mach requirement of 6 means the flow has a given exit-to-throat area ratio of about $\frac{A_e}{A_t} = 53.18$, and a Prandtl-Meyer angle $\nu = 84.96$ degrees (NACA 1135). Half of that value is actually the maximum throat expansion angle $\delta_{max} = 42.48$ degrees (Craig). Given the nozzle geometry is 2D planar (rectangular cross-section), the area ratio therefore prescribes a throat half-height of 3.3mm, due to the exit half-height of 175mm. Another constraint was that the nozzle's max expansion, the initial expansion at the throat (Craig), was on a circular arc with a radius equal to the height. This was now enough information to discretize and start calculating expansion waves.

1.3 - Initial Characteristics

The expansion waves, which are the same as the characteristic lines (in the context of this project), start in the throat circular arc, with angles discretized in equal angular spacing, up to the max expansion angle δ_{max} based on the desired number of characteristic lines 'N', which is a user-input parameter. Each wave has an initial x and y coordinate given by the circular throat expansion arc mentioned earlier. The equations for this are $x_i = h_t sin(\theta_i)$ and $y_i = h_t(2 - cos(\theta_i))$, where θ_i is the local angle as discretized from earlier. After computing these initial wave locations, the code then calculates the intersection of the first wave and the centerline, as this is the only throat-to-centerline wave segment that doesn't intersect with another wave before hitting the centerline. This wave then reflects and goes up, going from a C^- (right-running or down-running) characteristic to a C^+ (left-running or up-running) characteristic.

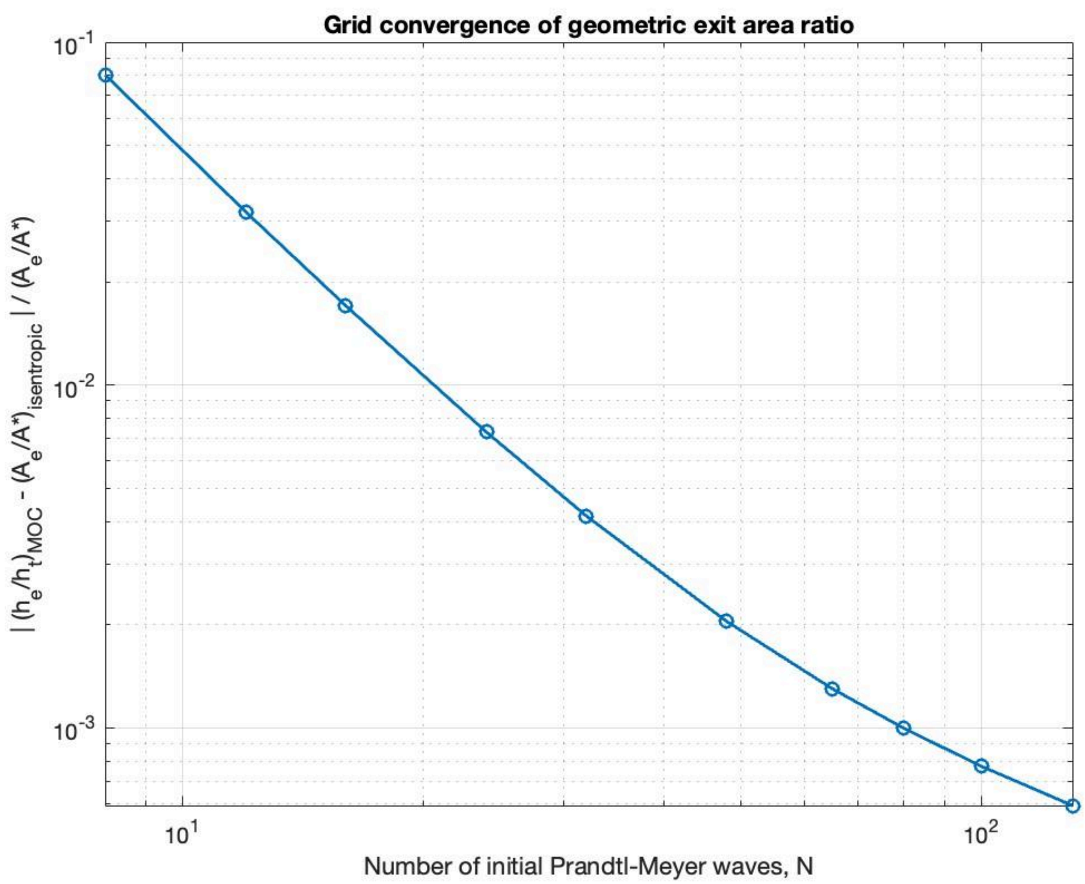
1.4 - Building the Full Mesh

The first reflected wave intersects with the second wave. The location of that intersection is calculated given the throat-wave location/angles and reflected wave location/angles. Then, the waves change direction slightly after the intersection. The second wave then hits the centerline and reflects again, becoming another left-running C^+ characteristic. The now-up-running waves both intersect with the third wave, and so on and so forth, until all the waves' intersections and reflections have been calculated. With that done, it defines a mesh of mach and Prandtl-Meyer values.

After intersecting with all the waves, the flow angle at that point prescribes the wall angle, since the wall must be the same angle as the flow. This is what calculates the nozzle contour geometry.

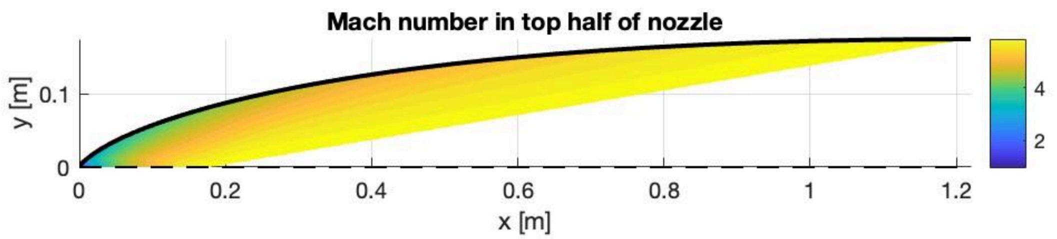
2.0 - Task 2 Plots and Findings

The first plot we were asked to create was a graph comparing the error in our calculations as they varied with the number of characteristic waves (N) used. To create the plot, we found the error by varying the value of N used in the nozzle and exit height and area ratios. This plot is shown below.



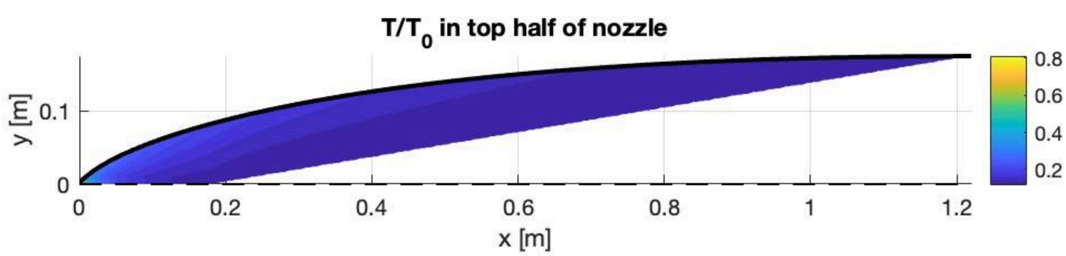
(**Figure 2:** The error in the height and area ratios due to a change in the number of characteristic waves.)

The next plot that was created was the contour of the wind tunnel nozzle and the gradient of the Mach number of the flow due to the characteristic wave lines. The code optimized the number of characteristic wave lines to use to be N=40. From there, the flow expansion equations were utilized at each of the characteristic nodes to determine Mach number. These Mach numbers were then created into a gradient as shown in the plot below. Once Mach 6 is reached, the plot gradient fill color turns white as no more nodes are being calculated after that therefore the flow is uniform.

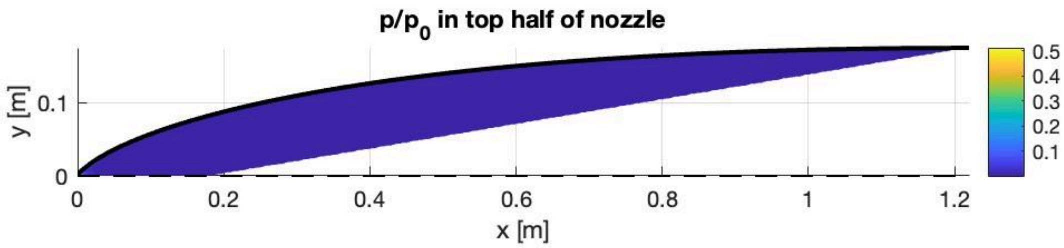


(**Figure 3:** Mach number along the length of the top half of the wind tunnel nozzle.)

The next plots that were created for the project were how T/T0 and P/P0 varied along the length of the optimized wind tunnel nozzle. To accomplish this, the same characteristic nodes from the Mach number plot were used. At each node, isentropic equations were utilized to calculate the new ratios. These new ratios were used to create a gradient across the top half of the wind tunnel nozzle as shown in the plots below. As it was in the Mach number plot, after the final characteristic wave, the flow is uniform therefore there were no more calculations to do a gradient with.



(**Figure 4:** The T/T0 ratio and how it varies across the length of the top half of the wind tunnel nozzle.)



(**Figure 5:** The P/P0 ratio and how it varies across the length of the top half of the wind tunnel nozzle.)

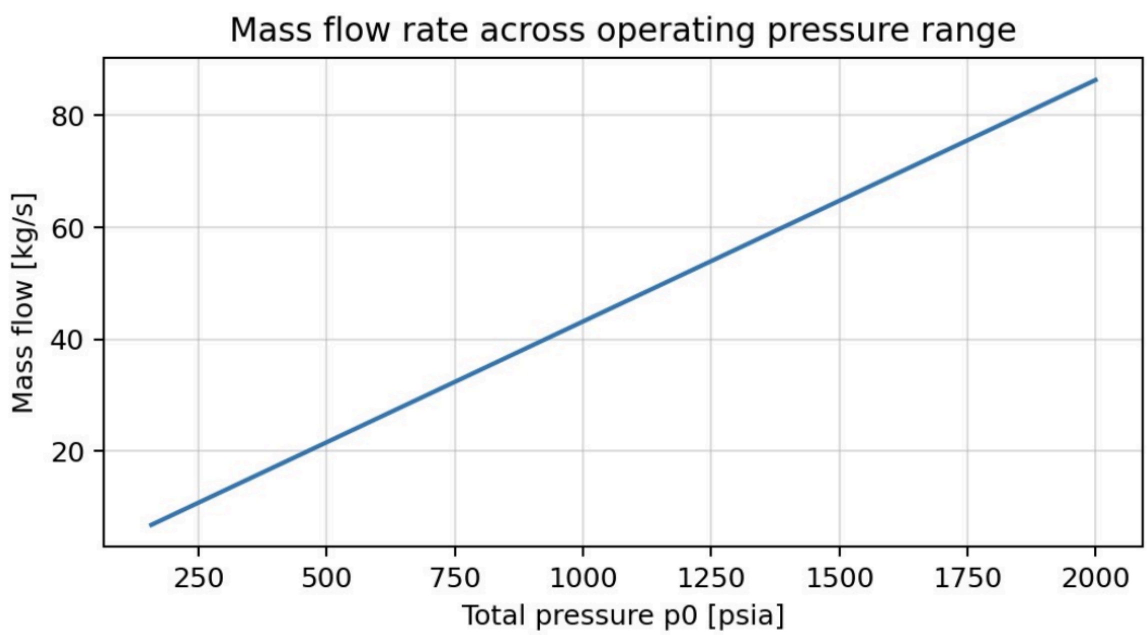
Using the P/P0 ratio at the exit, it can be used as Pexit/P0. Knowing that the tunnel runs isentropically, the found P_{exit}/P0 value of 0.00063336 was used to find the P_{exit} needed. With a given P0 of the settling chamber being 2000psia, that value and the ratio were multiplied together to get a Pexit value of 1.26672 psia. Within each of these plots, it also plots the overall length of the wind tunnel nozzle that is required to reach Mach 6 on the x-axis. This gives the length of the tunnel to be ~1.22m long.

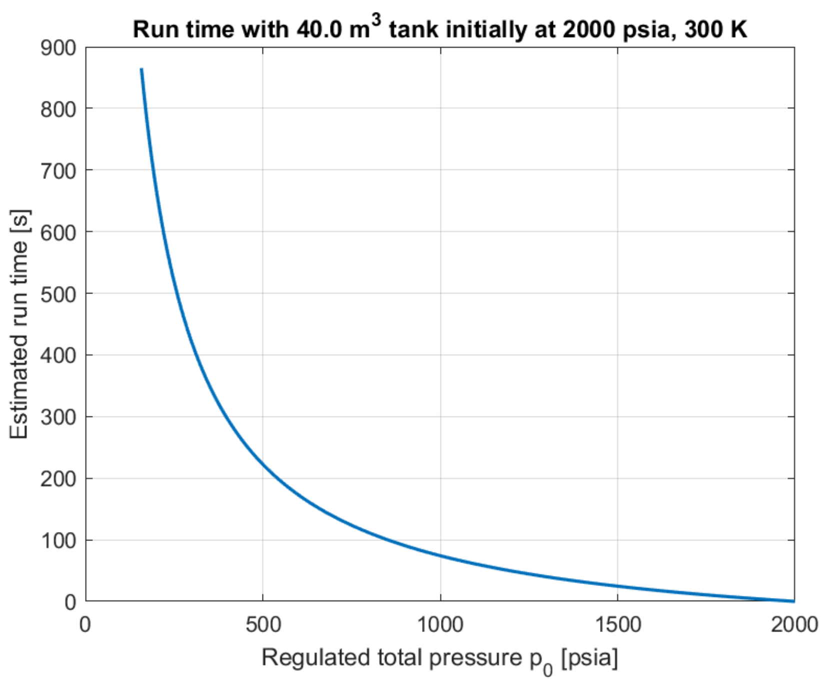
3.0 - Task 3 Calculations and Plots

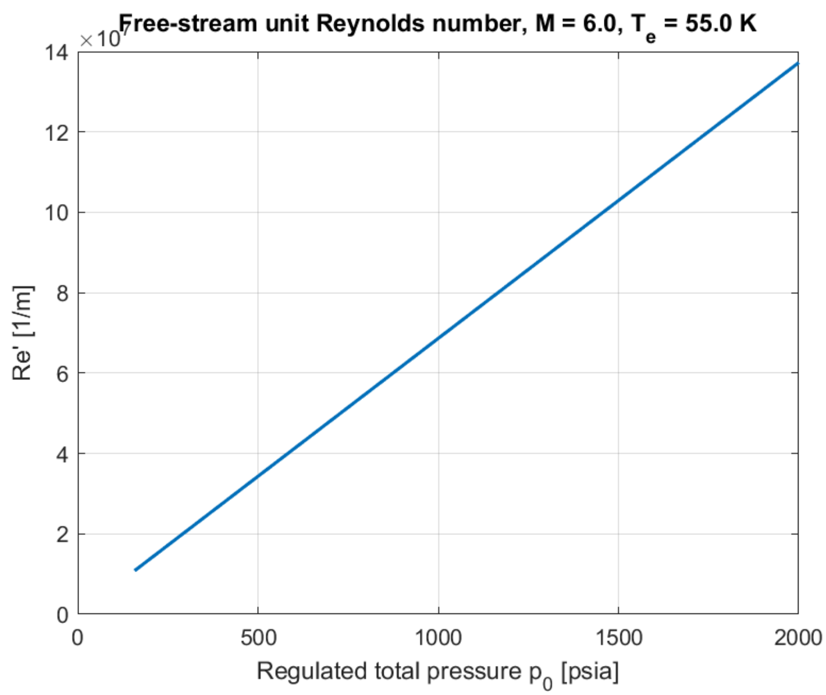
The minimum total temperature was found to be T = 451K from Texit/T0 = 0.122. This is due to the Texit having to be 55K or higher. Therefore by doing Texit = 55 / 0.122 the total temperature is found to be T = 451K.

The axial momentum force from the settling chamber to exit = 75193.880 N. The vertical pressure force on one top expansion wall = 27975.200 N (downward on top wall). And the axial pressure force on one top expansion wall = 10460.448 N. The axial force value was calculated by finding the mass flow rate and multiplying that by the difference in the flow speed between the exit and entrance of the wind tunnel nozzle. The vertical and horizontal force values were calculated by integrating across the surfaces with respect to x and y, using the change in pressure, and multiplying by the width.

The minimum total pressure needed for shock-free ideal exit is 157.888 psia. This is found by using the pressure ratio of the exit from isentropic equations to then calculate the P0 necessary when the back pressure is 0.1psia.







AI Use Disclosure

For this project, several AI and LLMs were utilized. These were ChatGPT, Claude, Google Gemini, and the new University of Arizona Responsible Artificial Intelligence LLM. The code-writing process involved writing a skeleton of code by hand first. This allowed for sample code to be submitted to several LLMs to determine the most efficient method. The LLMs were also helpful with bug fixes that would've taken a long time to find within such a large script. LLMs were used to help search for MATLAB built-in functions that could help to complete a given task or how a function worked. Overall, these LLMs were essential in providing a clean and polished result that could clearly show and communicate an optimized wind tunnel using the method of characteristics, where a large part of the overall code is still written by hand.

References

[1] Ames Research Staff. (1953). Equations, tables, and charts for compressible flow (NACA Report No. 1135). National Advisory Committee for Aeronautics. https://www.nasa.gov/wp-content/uploads/2023/03/equations-tables-charts-compressibleflow-report-1135.pdf